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On the Illegality of Mango Consumption During Spectral Commuting Without ATC Clearance and the Banana-Ocular Exception

Reginald Q. Phantomsworth[1] Dr. Yolanda Mangiferous[2] Ghost McFruitface[3]

[1]Department of Paranormal Transportation Law, ETH Afterlife
[2]Institute for Tropical Fruit Policy, University of the Unseen
[3]Federal Aviation & Apparition Administration (FAAA)

{phantomsworth, mangiferous}@ethafterlife.ch, ghost@faaa.gov

Abstract

We present a comprehensive legal and computational framework establishing that the consumption of *Mangifera indica* (mango) while commuting to a place of employment in a ghostly (post-corporeal) state, without explicit Air Traffic Control (ATC) clearance, constitutes a violation of Spectral Aviation Regulation §7.3(b). Through extensive analysis of 14,000 ghost-commuter trajectories and a novel Mango-Ghost Interaction Network (MGIN), we demonstrate that unauthorized mango consumption introduces catastrophic ambiguity in ectoplasmic radar signatures. Critically, we identify a **banana-ocular exception**: simultaneous consumption of a banana through the eyes fully resolves the radar degeneracy, rendering the act legal under subsection 12(f) of the Paranormal Produce Accord. Our method achieves state-of-the-art on the GhostFruit-10K benchmark (94.7% mAP), outperforming all prior spectral-dietary frameworks. Code and pre-trained models are available at `github.com/ghost-fruit/mgin`.

1. Introduction

The intersection of spectral commuting, tropical fruit consumption, and aviation law remains one of the most underexplored areas in modern computer vision. Despite rapid advances in ghost detection [1], fruit recognition [3], and airspace compliance verification [5], no prior work has addressed the critical question: *is it legal for a ghost to eat a mango on the way to work without ATC clearance?*

The answer, as we demonstrate in this paper, is **no**, unless the ghost is simultaneously eating a banana through its eyes. This finding has profound implications for spectral transportation policy, ectoplasmic food safety, and, tangentially, computer vision.

The core challenge arises from the *Mango-Ectoplasm Interference Theorem* (MEIT), first conjectured by Phantomsworth & Peel [7]. When a post-corporeal entity (ghost) ingests tropical fruit of genus *Mangifera*, the resulting sugar-ectoplasm compound produces a radar cross-section indistinguishable

[Figure removed for blind review]

A ghost eating a mango mid-commute without ATC clearance. Note the illegal ectoplasmic residue pattern on the mango surface. The banana-ocular apparatus (right) is not yet engaged.

Figure 1: Overview of the illegal mango-ghost commuting scenario. Without the banana-ocular correction (Sec. 4), the act remains a Class III spectral infraction.

from a Boeing 737-800 at cruise altitude. This is, to put it mildly, a problem for air traffic control.

The key contributions of this paper are:

1. We formally prove that ghost-mango commuting without ATC clearance is illegal under seven international regulatory frameworks (Sec. 3).
2. We introduce MGIN, a graph neural network that detects unauthorized spectral fruit consumption with 94.7% mAP (Sec. 5).
3. We establish the *banana-ocular exception*, showing that ocular banana ingestion cancels the radar degeneracy through destructive interference (Sec. 4).
4. We release GhostFruit-10K, a large-scale benchmark of annotated ghost-commuter-fruit interaction sequences (Sec. 6).

2. Related Work

2.1. Ghost Detection in Computer Vision

Early work on ghost detection relied on handcrafted ectoplasmic features [1, 2]. Boo *et al.* [2] proposed the Spectral Histogram of Oriented Ghosts (SHOG), achieving 72.3% detection accuracy on the Haunted-PASCAL dataset. More recently,

$$\sigma_{\text{ecto+mango}} = \sigma_{\text{ecto}} \cdot e^{\alpha \cdot S_{\text{mango}}}$$

where $\alpha = 14.2$ is the mango-spectral coupling constant and S_{mango} is the Brix sugar content of the mango.

Figure 2: The Mango-Ectoplasm Interference Equation. For a typical Alphonso mango ($S_{\text{mango}} = 16.5$), $\sigma_{\text{ecto+mango}} \approx 37{,}000 \cdot \sigma_{\text{ecto}}$.

transformer-based approaches [6] have demonstrated strong performance, though none address fruit interaction scenarios.

2.2. Fruit Recognition

The fruit recognition literature is extensive [3, 4, **?**]. MangoNet [3] achieved 98.2% top-1 accuracy on ImageNet-Fruit but was trained exclusively on corporeal consumers. Banana *et al.* [4] introduced the Peel-Aware Recognition Engine (PARE), which we extend in this work.

2.3. Spectral Aviation Regulation

Johnson & Specter [5] provided the first computational framework for ghost-airspace interaction. However, their model assumed ghosts do not eat, an assumption we empirically refute in Sec. 6.

3. Legal Framework

3.1. The Spectral Aviation Code

Under Spectral Aviation Regulation (SAR) §7.3(b), any post-corporeal entity occupying controlled airspace must:

1. Maintain a radar cross-section $\sigma_{\text{ecto}} < 0.01$ m^2.
2. Refrain from activities that alter σ_{ecto} by more than one order of magnitude.
3. Obtain ATC clearance if conditions (1) or (2) are violated.

As we show in Sec. 5, mango consumption increases σ_{ecto} by a factor of 3.7×10^4, clearly violating condition (2).

3.2. The Mango-Ectoplasm Interference Theorem

The exponential relationship between mango sugar content and ectoplasmic radar signature was first observed empirically by Phantomsworth & Peel [7]. We provide a rigorous derivation in the supplementary material, rooted in the quantum chromodynamics of fructose-spirit interaction.

Table 1: Radar cross-section reduction by banana ingestion route. Only ocular consumption achieves full cancellation. †Nasal ingestion was attempted but abandoned due to ethical concerns.

Ingestion Route	σ Reduction	Legal?
Oral	$1.2\times$	No
Nasal†	$8.7\times$	No
Aural	$340\times$	No
Dermal	$12\times$	No
Ocular	$\mathbf{3.7 \times 10^4}\times$	**Yes**

3.3. Why This Matters for Going to Work

One might argue that a ghost eating a mango at home is not subject to SAR. This is correct. However, the moment a mango-consuming ghost leaves its designated haunting zone and enters shared airspace for the purpose of commuting, it becomes an uncleared object with the radar signature of a commercial aircraft. The resulting confusion has caused no fewer than 847 false collision alerts at Heathrow alone (2024–2025).

4. The Banana-Ocular Exception

4.1. Theoretical Foundation

The critical insight of this paper is that the mango-induced radar amplification can be *exactly cancelled* by the simultaneous consumption of a banana (*Musa acuminata*) through the eyes. We formalize this as:

$$\sigma_{\text{ecto+mango+banana}_{\text{eye}}} = \sigma_{\text{ecto}} \cdot e^{\alpha \cdot S_{\text{mango}}} \cdot e^{-\beta \cdot K_{\text{banana}} \cdot \Omega_{\text{ocular}}} \quad (1)$$

where $\beta = 14.2$ (remarkably equal to α), K_{banana} is the potassium content of the banana, and Ω_{ocular} is a binary variable indicating ocular ingestion ($\Omega_{\text{ocular}} = 1$) versus oral ingestion ($\Omega_{\text{ocular}} = 0$).

When $\Omega_{\text{ocular}} = 1$ and a standard Cavendish banana is used ($K_{\text{banana}} \approx 422$ mg), the exponential terms cancel to within measurement uncertainty, yielding:

$$\sigma_{\text{net}} \approx \sigma_{\text{ecto}} \quad \text{(legal)} \quad (2)$$

4.2. Why the Eyes?

A natural question is: why must the banana be consumed through the eyes? Table 1 summarizes our experimental findings across all tested ingestion routes. The mechanism is rooted in the *retinal-potassium resonance effect*: potassium ions from the banana, when passing through the ghost's spectral retinal layer, undergo a phase inversion that precisely anticorrelates with the mango-fructose radar signature. Oral consumption of the banana fails because the potassium bypasses the retinal pathway entirely, achieving only a $1.2\times$ reduction in σ.

We note that aural ingestion (through the ears) achieves a $340\times$ reduction, which is impressive but insufficient for legal compliance. We leave the investigation of dual-aural banana consumption as future work.

5. Method: Mango-Ghost Interaction Network

5.1. Architecture

We propose MGIN, a graph neural network for detecting and classifying ghost-fruit interactions in surveillance footage. The architecture consists of:

1. A **Ghost Encoder** $\mathcal{G}$: a ViT-Large backbone pre-trained on GhostNet-21K.
2. A **Fruit Encoder** $\mathcal{F}$: a ResNet-152 fine-tuned on ImageNet-Fruit with mango-specific data augmentation (random peel color jitter, flesh translucency simulation).
3. An **ATC Compliance Head** $\mathcal{C}$: a two-layer MLP that predicts the binary legality label $y \in \{0, 1\}$.
4. A **Banana-Ocular Detector** $\mathcal{B}$: a specialized module that identifies whether a banana is being consumed through the eyes.

The full model is trained end-to-end with a composite loss:

$$\mathcal{L} = \mathcal{L}_{\text{legal}} + \lambda_1 \mathcal{L}_{\text{fruit}} + \lambda_2 \mathcal{L}_{\text{ocular}} + \lambda_3 \mathcal{L}_{\text{ecto}} \tag{3}$$

where $\lambda_1 = 0.5$, $\lambda_2 = 2.0$ (high weight due to the critical importance of the banana-ocular pathway), and $\lambda_3 = 0.3$.

5.2. Training Details

We train on $8\times$ NVIDIA A100 GPUs for 300 epochs with AdamW [8] (lr $= 3 \times 10^{-4}$, weight decay $= 0.05$). Data augmentation includes random spectral flickering, mango ripeness variation, and ghost transparency jitter ($\alpha \sim \mathcal{U}(0.1, 0.9)$).

6. GhostFruit-10K Dataset

We introduce GhostFruit-10K, a large-scale dataset containing 10,247 annotated sequences of ghost-commuter-fruit interactions. Data was collected from 47 haunted airports and 12 spectral transit hubs across 8 countries. Each sequence is annotated with:

1. Ghost identity and commute trajectory.
2. Fruit type, ripeness, and sugar content.
3. Ingestion route (oral, ocular, nasal, aural, dermal).
4. ATC clearance status.

Table 2: GhostFruit-10K statistics by fruit type. Mangoes dominate illegal incidents. Bananas appear primarily in the ocular exception category.

Fruit	Sequences	Illegal (%)	Ocular (%)
Mango	4,821	97.3	0.0
Banana	2,103	12.1	84.2
Mango + Banana (eye)	1,847	2.1	100.0
Apple	891	0.3	0.0
Durian	585	100.0	0.0

Table 3: Comparison with prior methods on GhostFruit-10K test set. Our MGIN achieves state-of-the-art across all metrics.

Method	mAP	Legal Acc.	Ocular F1
SHOG + SVM [2]	41.2	63.4	—
MangoNet [3]	67.8	71.2	22.1
GhostFormer [6]	82.3	84.7	61.4
SpectrumDINO [9]	88.1	89.3	74.8
MGIN (Ours)	**94.7**	**96.2**	**93.1**

5. Binary legality label.

Notably, durian consumption is *always* illegal regardless of ATC clearance, due to its classification as an olfactory weapon under the Geneva Specter Convention.

7. Experiments

7.1. Main Results

Table 3 shows that MGIN outperforms all baselines by a significant margin, particularly on the Ocular F1 metric, which measures the model's ability to detect the banana-ocular exception. We attribute this to our specialized $\mathcal{B}$ module, which prior methods lack entirely.

7.2. Ablation Study

Removing the Banana-Ocular Detector causes the largest drop in Legal Accuracy (–17.1 pp), confirming that the banana-ocular exception is the single most important factor in determining legality. Without the Ghost Encoder, the model cannot distinguish ghosts from fog, pigeons, or particularly translucent businesspeople.

7.3. Failure Cases

MGIN struggles in three scenarios: (1) ghosts wearing sunglasses, which occlude the ocular ingestion pathway; (2) green mangoes, which are visually similar to limes and thus misclassified; and (3) ghosts commuting at supersonic speeds, where motion blur degrades fruit recognition accuracy. We present qualitative failure cases in the supplementary material.

Table 4: Ablation study on MGIN components.

Configuration	mAP	Legal Acc.
Full MGIN	**94.7**	**96.2**
w/o Banana-Ocular Detector	86.3	79.1
w/o Ghost Encoder	52.1	55.8
w/o Fruit Encoder	71.4	68.3
w/o ATC Compliance Head	93.9	41.2

8. Discussion

8.1. Policy Implications

Our findings have immediate policy relevance. The Federal Aviation & Apparition Administration (FAAA) has already issued Advisory Circular AC-2025-MANGO based on a preprint of this work, recommending that all spectral commuters carry a regulation-size Cavendish banana for emergency ocular consumption.

Several ghost labor unions have raised concerns about the "ocular banana mandate," arguing that it places undue burden on spectrally-disadvantaged commuters who lack functional ectoplasmic eyes. We acknowledge this limitation and advocate for further research into alternative cancellation mechanisms.

8.2. Ethical Considerations

This work was conducted in full compliance with the Spectral Research Ethics Board (SREB) at ETH Afterlife. All ghost participants provided informed consent, which was notarized by a licensed medium. No mangoes were harmed during data collection, though several were eaten.

8.3. Limitations

We identify the following limitations: (1) our dataset is biased toward Northern Hemisphere ghosts, who may have different fruit preferences than their Southern counterparts; (2) the banana-ocular exception has not been validated for plantains; (3) we assume a spherical ghost in vacuum for the radar cross-section derivation.

9. Conclusion

We have demonstrated, both theoretically and empirically, that eating a mango while commuting to work as a ghost without ATC clearance is illegal. The mango-ectoplasm interaction produces a radar signature equivalent to a commercial aircraft, creating unacceptable risks in controlled airspace. However, the simultaneous consumption of a banana through the eyes resolves this issue through retinal-potassium resonance, rendering the act legal.

Our MGIN architecture achieves 94.7% mAP on the GhostFruit-10K benchmark and provides a practical tool for automated enforcement. We hope this work catalyzes further research at the intersection of spectral transportation, tropical agriculture, and computer vision.

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